The FIFO linking theorem: affine response and the cut-space frontier

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August 2026

Abstract

This note attacks the general linking theorem underlying the ECHO-FIFO+dummy Gold-quadric realizer. A finite graph is played by opening vertices into a FIFO queue and later closing the front. A close flips a bit when the front has odd degree into the untouched set. The target theorem says that one isolated dummy lets the even player force even total flip parity from either seat, for every graph.

The theorem is not proved here. The contribution is an exact algebraic reduction that replaces the failed local partner/debt induction by a global affine-flow target. Every play has a vector-valued score: its graph of disjoint activity intervals. For every fixed attacker strategy, the linking theorem is equivalent to saying that zero lies in the $\mathbb{F}_2$-affine hull of all response scores. At a real FIFO front f, the edge space splits canonically as

$$E(K_L) = \mathrm{Cut}(K_L) \oplus E(K_{L\setminus\{f\}}),$$

so openings before f closes occupy one cut-space layer and all later play occupies the smaller edge-space layer. This gives a canonical filtration, but not a branchwise induction: an exact $P_4 + d$ state has defender children whose projected continuation affine hulls are disjoint.

Two positive theorems accompany the reduction. The vector live-star identity proves the affine formulation, and a parity-cell pairing theorem strictly extends the earlier twin-pair solution. Exact small witnesses show why neither same-degree pairing nor greedy FIFO rotation closes the general case. The remaining obligation is a global odd-flow contraction which couples branches across the filtration rather than solving each projected child separately. The note therefore advances the proof frontier without promoting the complete finite census through eight real vertices to a proof.

1 The reduced game

Let $G = (R, E)$ be a finite simple graph on the *real* vertices, and adjoin one isolated vertex d. A state is (U, Q, ko), where U is the set of untouched vertices, Q is an ordered FIFO queue of open vertices, and ko records the one-move prohibition after a vertex is opened onto an empty queue. A legal move is

1. OPEN(x) for any $x \in U$, moving x to the back of Q; or
2. CLOSE(f) for the front f of Q, unless ko is set.

A close of f scores

$$\deg_U(f) \pmod 2.$$

If no move is legal, the forced pass only clears ko. This can occur only after U is empty, when no later close can score. The players have opposite targets for the parity of the total score.

[PROVED] Every vertex opens and closes exactly once. FIFO makes the opening and closing orders identical. Consequently the activity intervals $I_x = [o(x), c(x)]$ never properly nest. For an edge xy with $o(x) < o(y)$, the edge contributes to the flip score precisely when $c(x) < o(y)$, i.e. precisely when I_x and I_y are disjoint. Thus the total score is

$$F_G(h) = \sum_{xy \in E(G)} [I_x \cap I_y = \varnothing] \pmod 2. \tag{1}$$

The proper-interval vocabulary is standard; the adversarial parity problem is not a standard queue-layout problem, because the queue stores vertices and the play itself constructs their intervals.

2 The vector score

Write

$$\mathcal{E}_R = \mathbb{F}_2^{\binom{R}{2}}$$

for the vector space whose coordinates are unordered pairs of real vertices. For a completed history h, define its *disjointness vector*

$$D(h) = \sum_{\{x,y\} \subseteq R} [I_x \cap I_y = \varnothing], xy \in \mathcal{E}_R.$$

Pairing $D(h)$ with the adjacency vector A_G recovers (1).

At any moment let $L = U \cup Q$ be the live set. For a real $x \in L$, put

$$s_L(x) = \sum_{y \in (L \cap R) \setminus \{x\}} xy,$$

and set $s_L(d) = 0$. These are the vertex stars, or simplicial coboundaries, of the live complete graph.

Theorem 1 (vector live-star identity)**.** **[PROVED]** *Every completed history satisfies*

$$D(h) = \sum_{x \text{ opened}} s_{L_x}(x), \tag{2}$$

where L_x is the live set at the instant x opens.

Proof. Fix a real pair xy and suppose x opens first. The vertex y is live when x opens, so xy occurs once in $s_{L_x}(x)$. If the activity intervals overlap, x is still live when y opens and the coordinate occurs a second time in $s_{L_y}(y)$; the two occurrences cancel over $\mathbb{F}_2$. If the intervals are disjoint, x has already closed when y opens and there is no second occurrence. The coordinate of the right-hand side is therefore exactly the coordinate of $D(h)$. □

Corollary 2 (scalar live-degree identity)**.** **[PROVED]** *For every graph G,*

$$F_G(h) = \sum_{x \text{ opened}} \deg_{G[L_x]}(x) \pmod 2.$$

This upgrades the earlier scalar potential calculation to an identity before a graph is chosen. The graph enters only through the final linear functional $z \mapsto \langle A_G, z \rangle$.

3 Affine response is exactly the theorem

Fix which seat seeks odd parity, and fix a deterministic strategy S for that attacker. Move legality is graph-independent, so S can be regarded as a strategy in the schedule game before an adjacency vector is chosen. Let $\mathcal{H}_S$ be the set of terminal histories compatible with S and with arbitrary legal responses by the other player.

Theorem 3 (affine separation)**. [PROVED]** *The strategy S forces odd score for some graph on R if and only if*

$$0 \notin \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\}.$$

Consequently, the general linking theorem is equivalent to the following graph-free statement for both choices of attacker seat:

$$\textit{for every } S, \qquad 0 \in \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\}. \tag{3}$$

Proof. If S forces odd score for a graph with adjacency vector A, then $\langle A, D(h)\rangle = 1$ for every $h \in \mathcal{H}_S$. The whole affine hull therefore lies in the affine hyperplane $\langle A, z\rangle = 1$, which excludes zero.

Conversely, write the affine hull as $a + W$. If it excludes zero, then $a \notin W$. Elementary linear separation over $\mathbb{F}_2$ supplies a linear functional ℓ with $\ell(W) = 0$ and $\ell(a) = 1$. Every linear functional on $\mathcal{E}_R$ is pairing with the adjacency vector of a simple graph on R. For that graph, $\ell(D(h)) = 1$ for every response history, so S forces odd score. □

Equivalently, one must select an odd-cardinality formal multiset of response histories whose disjointness vectors XOR to zero. This is an $\mathbb{F}_2$-flow problem: attacker nodes route all incoming weight along the move prescribed by S, while defender nodes may split it among an odd number of outgoing moves. Uniformly summing all response leaves does *not* work in general; the flow must be selected.

There is a useful exact homogeneous recursion. At a history h, let $Z(h) \subseteq \mathbb{F}_2 \oplus \mathcal{E}_R$ be the span of $(1, D_h(\lambda))$ over response leaves λ below h, where D_h is the future disjointness increment. Give a close of a real front f the charge $c = \sum_{u \in U \cap R} fu$, and give an open, dummy close, or pass charge zero. This sums to $D(h)$ because a pair becomes disjoint exactly when its earlier vertex closes while the later one is untouched. Put

$$\tau_c(a, z) = (a, z + ac).$$

Then a terminal node has $Z = \mathrm{span}\{(1, 0)\}$; an attacker node, unary after fixing S, has $Z(h) = \tau_c Z(h')$; and a defender node has

$$Z(h) = \mathrm{span}_m \, \tau_{c_m} Z(h_m).$$

The desired affine response is exactly $(1, 0) \in Z(\mathrm{root})$. Thus the open problem is a label-preserving contraction of this recursion, not an ordinary contraction of the response tree: ordinary boundaries have even augmentation, while the required leaf chain has odd augmentation.

[TESTED] The existing exhaustive graph census through eight real vertices plus dummy, both attacker seats, implies (3) in those dimensions by Theorem 3. Independent graph-free policy searches also found no affine offset through eight real vertices; selected structured policies at nine and ten real vertices likewise had zero offset. These are finite checks, not a proof of (3).

4 A solved parity-cell subclass

The true/false-twin argument in the earlier draft can be weakened substantially.

Theorem 4 (parity-cell pairing)**.** **[PROVED]** *Suppose all vertices, including the dummy if present, admit a perfect partition $\mathcal{M}$ into two-element cells such that*

$$e_G(A,B)=0 \pmod 2 \qquad \textit{for every distinct } A,B\in\mathcal{M}. \tag{4}$$

Then the second player forces even flip parity.

Proof. When the first player opens one member of a cell A, the second opens its mate. When the first player closes the FIFO-front member of A, its mate is the next queue front and the second closes it. Thus untouched vertices are always a union of whole cells, and the queue is a concatenation of whole cells. Ko causes no exception because the forced second opening completes the first cell.

When the two members a, a' of A close consecutively, their two flip bits sum to

$$\deg_U(a)+\deg_U(a')=\sum_{B\subseteq U} e_G(A,B)=0$$

by (4). Every flip is contained in one such consecutive pair, so the total parity is even. □

Twin pairs satisfy (4), but the condition only asks for even cross-parity between cells; no automorphism is required.

[TESTED] This theorem is still not general. Among the 1,044 isomorphism classes on seven real vertices, 64 boards obtained by adjoining the isolated dummy have no such partition. The first witness in the local ordering has real edges

$$02, 04, 06, 14, 15, 23.$$

The general linking theorem nevertheless passes exact minimax on every one of those boards. Seidel switching cannot repair the pairing obstruction: between two two-cells A, B, switching a set S toggles

$$|A\cap S||B\setminus S|+|A\setminus S||B\cap S|\equiv ab+ab=0 \pmod 2.$$

5 FIFO gives a cut-space filtration

Let $L\subseteq R$ be a live real set and let $f\in L$ be the FIFO front. Let $E(L)=\mathbb{F}_2^{\binom{L}{2}}$ and let

$$\mathrm{Cut}(L)=\mathrm{span}\{s_L(x): x\in L\}.$$

This is the cut space of the complete graph on L.

Theorem 5 (front splitting)**.** **[PROVED]** *There is a canonical direct sum*

$$E(L)=\mathrm{Cut}(L)\oplus E(L\setminus\{f\}). \tag{5}$$

The projection to the second summand is

$$(T_f z)_{ij}=z_{ij}+z_{fi}+z_{fj}, \qquad i,j\neq f. \tag{6}$$

It kills every live star and is the identity on the canonical $E(L\setminus\{f\})$ summand.

Proof. For $x = f$, the two incident coordinates in (6) cancel; for $x = i \neq f$, the ij and fi coordinates of $s_L(i)$ cancel. Hence T_f kills all generators of $\mathrm{Cut}(L)$. If z is supported away from f, then $T_f z = z$. Finally,

$$\dim \mathrm{Cut}(L) + \dim E(L \setminus \{f\}) = (|L| - 1) + \binom{|L|-1}{2} = \binom{|L|}{2},$$

so the two subspaces are complementary and exhaust $E(L)$. □

While f remains front, opens do not change the live set. By Theorem 1, all their contributions lie in the current $\mathrm{Cut}(L)$ layer. Once FIFO deletes the real front f, every later contribution is supported on $E(L \setminus \{f\})$. Repeating the splitting along the forced closure order gives a triangular decomposition of the complete play score. This is the missing structural use of FIFO: the earlier scalar potential was valid even if any queued vertex could close and therefore could not prove the theorem.

The dummy needs a companion state in this induction. Its star is zero, but closing it does not shrink the real edge space. A proof must use that zero star to adjust affine augmentation and then transfer the odd response flow to the unchanged real live set. Treating the dummy as an ordinary deleted front silently loses precisely the exceptional case.

6 Why the deterministic local induction fails

The live-degree pairing heuristic is correct only conditionally. At a turn of the block initiator the queue has even length. If the initiator opens x, the responder would like to open an untouched y with the matching live-degree parity. If no such y exists and the old queue is nonempty, closing the old front rotates

$$(a_1, b_1, \dots, a_r, b_r, x) \longmapsto (b_1, a_2, \dots, a_r, b_r, x).$$

Neither operation is a complete strategy.

[TESTED] The smallest useful witnesses are exact minimax states, not counterexamples to the linking theorem.

1. On $K_2 + d$, after

   $$A : O_d, \qquad D : O_0, \qquad A : C_d,$$

   the defender faces $Q = (0)$ and $U = \{1\}$. The winning move is O_1; the apparently natural mirror close C_0 loses.

2. On the real path $3 - 0 - 1 - 2$ plus d, after

   $$A : O_0, \qquad D : O_1, \qquad A : O_d,$$

   the defender faces $Q = (0, 1, d)$ and $U = \{2, 3\}$. Both available opens have the wrong live-score parity, and closing 0 loses. The unique winning move is the proactive wrong-parity open O_3, which creates an even odd-front corridor and stores a debt before the local trap appears.

Thus a state descriptor consisting only of current debt, front parity, parity classes in U, and one FIFO rotation is insufficient. The missing datum is a recursive repair certificate in the untouched graph. This is exactly what the affine flow is allowed to encode globally.

7 The exact local residue

The scalar cut potential gives a complete local classification, provided its limitation is stated explicitly. Regard the queue as a set when counting edges and put

$$P(Q, U) = e_G(Q, U) \pmod 2.$$

For a move m, let $\chi(m)$ be the change in P. If $R = Q \cup U$, direct edge accounting gives

$$\chi(O_x) = \deg_{G[R]}(x), \qquad \chi(C_f) = \deg_U(f), \qquad \chi(\text{pass}) = 0.$$

The middle term is the actual flip.

Proposition 6 (charge-matching and the poison state)**.** **[PROVED]** *Suppose $P = 0$, $|U|$ is even, and the position is nonterminal. After an opponent open, there is always a distinct legal open with the same χ. If a reply is due after an opponent close of charge c, there is a legal response of charge c except in exactly the following case:*

$$c = 0, \quad U \neq \varnothing, \quad \deg_{R'}(u) = 1 \pmod 2 \textit{ for every } u \in U, \quad Q' = \varnothing \textit{ or } \deg_U(\text{front}\, Q') = 1 \pmod 2, \tag{7}$$

where $Q' = Q \setminus \{f\}$ and $R' = Q' \cup U$ after the close.

Proof. At $P = 0$,

$$\sum_{u \in U} \deg_R(u) = e(U, Q) = 0 \pmod 2,$$

because edges internal to U are counted twice. Hence the odd-charge opens form an even set; since $|U|$ is even, the even-charge opens do too. The charge class containing the opponent's opener therefore contains another vertex. Opening never has a ko obstruction, and the live set R is unchanged by an open, so this is a legal matching response.

If the opponent closes f with charge c, then

$$\sum_{u \in U} \deg_{R'}(u) = e(U, Q') = c \pmod 2.$$

For $c = 1$ an odd-charge open exists. For $c = 0$ an even-charge open exists unless every untouched vertex has odd charge; in that exceptional case a matching response still exists precisely when the new FIFO front exists and has even degree into U. A close clears ko, so that front close is legal. This is exactly the complement of (7). When U is empty all remaining closes, including the close after the unique terminal pass, have charge zero. □

Matching χ restores $P = 0$, but does *not* by itself pair the flip score: a mixed odd-close/odd-open round contains one actual flip. This is the debt which the global chain must transport. The dummy prevents (7) while it is untouched (it is an even-charge open) or is the new front (it is an even close), but it may be buried in the queue or already closed. The proposition therefore isolates the local obstruction without pretending to solve its recurrence.

8 The remaining global lemma

The proof target can now be stated without Gold forms, graph enumeration, or a particular local menu.

Question 7 (global FIFO affine contraction). **[OPEN]** Fix either attacker seat and a deterministic attacker strategy S in the schedule game with one isolated dummy. Does there always exist an odd $\mathbb{F}_2$-flow through the defender branches whose terminal disjointness moment is zero, with the flow allowed to couple different defender children and different levels of the front filtration?

An affirmative answer proves the general linking theorem by Theorems 3 and 5. The tempting stronger claim—that projected continuation hulls of different defender children meet one common coset—is false. At the $P_4 + d$ state above, fix a history-dependent attacker which follows different continuations after O_2 and O_3. Under the front projection T_0 to the edge coordinates $12, 13, 23$, exact recursion gives the two open-child continuation hulls

$$\{0\}, \qquad \mathrm{Aff}\{23, 12 + 23\}.$$

They are disjoint. Exhaustive recursion over the smaller states with two real open children shows that this is the first such childwise failure.

Thus a valid contraction must transport nonzero target cosets and cancel them *across* defender children; it cannot solve every projected child as an independent zero-target subproblem. In particular, an induction classified only by dummy presence, live-set parity, attacker seat, and front position is false on reachable states. The local relations

$$s_L(d) = 0, \qquad \sum_{x \in L \cap R} s_L(x) = 0$$

provide the cone and the cut dependence, but not the required global coupling.

This boundary also explains why the nearest imported techniques do not close the problem. A standard strategy-stealing pass must be optional and harmless; the dummy has two mandatory events which change FIFO legality. A static copycat argument requires an involution or a parity-cell partition, absent on the 64 finite witnesses above. Local complementation changes the required pairing data but is not a legal FIFO move. Queue layouts supply terminology for nonnesting but no adversarial parity strategy.

[PROVED] The smallest genuine affine cancellation is already a three-history fan, not a dummy cone. With real vertices $1, 2, 3$, let a first-seat attacker open the dummy and thereafter close whenever possible, otherwise opening the least untouched vertex. After the shared prefix O_d, O_1, C_d, the defender can realize the three compatible terminal histories

	remaining moves	$D(h)$
(i)	O_2, C_1, O_3, C_2, C_3	$\{13\}$
(ii)	O_3, C_1, O_2, C_3, C_2	$\{12\}$
(iii)	C_1, O_2, O_3, C_2, C_3	$\{12, 13\}$.

Their three vectors XOR to zero, although none is zero. In (iii), O_2 opens an empty queue and O_3 is the ko-forced response, so all three rows are legal. The cancellation happens after the dummy has closed.

[TESTED] Even the natural fan over *all* second openings is too rigid. On real vertices $0, 1, 2$ plus d, let the attacker first play O_0; after replies O_1 or O_2 use "open least untouched, otherwise close", and after O_d use "close whenever possible, otherwise open least". Exact terminal affine hulls conditioned on the second reply, in coordinates $01, 02, 12$, are

$$\mathcal{A}_1 = \{0\}, \qquad \mathcal{A}_2 = \{0\}, \qquad \mathcal{A}_d = \{3, 7\}.$$

The immediate star charges of the three branches XOR to zero, but no one-leaf-per-branch transversal does. The global affine target survives by retaining only an odd subflow inside $\mathcal{A}_1$ or $\mathcal{A}_2$. Hence a proof must choose its odd subset of defender branches using continuation data; neither dummy-edge contraction nor the full root fan is a valid fixed recipe.

9 Consequence for the Gold realizer

[PROVED] The reductions in `writeups/goldarf.tex` identify the terminal σ-charge with the overlap parity of G-edges. Since overlap and disjointness partition $E(G)$,

$$\sigma(h) = |E(G)| + F_G(h) \pmod 2.$$

Therefore an affirmative answer to Question 7 would force $\sigma = |E(G)|$ for every support graph and both seats. On a Gold support graph this is the required quadratic value, upgrading the verified $m \in \{4, 8\}$ realizer to every m.

[OPEN] This would still solve only the mechanism half of the original `tis` problem. The realizer is σ-valued; recasting its forced charge as a natural normal-, misère-, or loopy-outcome P-set remains a separate open step.

10 Verification boundary

The following claims have different status and should not be flattened.

- **[PROVED]** The interval/disjointness reduction, vector live-star identity, affine-separation equivalence, homogeneous response recursion, parity-cell theorem, front cut-space splitting, charge/poison classification, and displayed triangular fan are complete proofs in this note.
- **[TESTED]** Exact minimax verifies the general linking theorem on every isomorphism class through eight real vertices plus dummy, both seats, and verifies the two local-strategy witnesses. The parity-cell census is complete through seven real vertices.
- **[TESTED]** The childwise common-coset strengthening is false, first at the displayed $P_4 + d$ state. This is a failure of the proposed induction, not of the linking theorem.
- **[TESTED]** The mandatory full second-opening fan also fails at three real vertices plus dummy, despite the immediate star identity. The global affine target survives inside a single reply branch.
- **[OPEN]** The global FIFO affine-contraction statement (3) is not proved. Neither the complete finite census nor the graph-free affine searches are evidence of a general chain contraction.

References

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